Supplier Watch — Appendix

Reference material for the submission document.

- A. Data dictionary
- B. Full DDL (PostgreSQL)
- C. Sample data as loaded
- D. Query results (PostgreSQL 17)

Appendix A — Data dictionary

Column-by-column reference for the 12 tables. The model overview is in Part A.

dim_site — one physical plant (Type 1)

Column	Description
site_key	Our own ID, e.g. S-001. Generated by us. Agencies never see it.
site_name	Display name we chose, e.g. "Acme Pharma Unit II, Sangareddy".
country_code	ISO-2.
city	Normalised city.
address	Normalised address. Used for matching and facets.
is_active	true normally; false once merged into another site. The target is visible in the bridge: the site's identifiers now point at the survivor, grouped by one decision_group_id.
created_at	When we created the site.

Type 1 because site *names* changing is not the history customers care about; the history that matters (which facts belonged to the site on date X) lives in the bridge.

dim_site_identifier — the bridge (SCD2)

One row = "identifier X belongs to site Y from date A to date B". A phone book from agency ID to our site.

Column	Description
identifier_type	FEI (FDA), EU_SITE_REF (EU), LETTER_ID (warning letter with no FEI, matched on its own), WATCH_ID (a customer's watchlist row).
identifier_value	The ID: 3004521877, LOC-118842, WL-320-26-118, 9912.
site_key	Which of our sites it points to.
match_method	EXACT_ID (shared agency ID), RULE (deterministic rule), FUZZY (name/address score), STEWARD (human decision).
match_score	0–1 for FUZZY; 1.0 for EXACT_ID and STEWARD.
match_evidence	JSON: normalised strings and rule that matched, so a steward can see *why*.
matching_rules_version	Which ER rule set produced the row. Lets us run v3 next to v2 and compare.
valid_from, valid_to, is_current	SCD2. A merge closes the old row and opens a new one with a different site_key.
decided_by	Who made this link: a steward's user ID, or system:er-v3 for automatic matches.
decision_reason	Free text from the steward (NULL for automatic matches).
decision_group_id	One UUID shared by every row a single steward action opened or closed. A SPLIT undoes a MERGE by finding all rows with the merge's group ID.

Key: `(identifier_type, identifier_value, valid_from)`. **Rule:** one identifier → exactly one site at any moment (enforced by a constraint).

fact_inspection — S1, FDA inspection classifications (SCD2)

Column	Description
`inspection_id`	FDA's ID, e.g. `IN-2041187`. Natural key.
`valid_from`, `valid_to`, `is_current`	SCD2; `valid_from` = when the weekly snapshot containing this version was loaded.
`record_status`	`ACTIVE` or `RETRACTED`. Set to `RETRACTED` when the row disappears from a snapshot. Row kept for audit.
`fei_number`	FDA's site identifier → `dim_site_identifier`.
`firm_name_as_published`	Verbatim from the CSV. Both Acme spellings are kept.
`city`	As published.
`country_code`	ISO-2, normalised from `India` / `IN`.
`inspection_end_date`	Parsed date.
`inspection_end_date_as_published`	Original text, kept because `08/02/2026` is ambiguous and the parser may change.
`classification`	`NAI` / `VAI` / `OAI`.
`product_type`	As published.
`published_date`	As published.
`batch_id`	Which weekly file this version came from → `ingest_batch.s3_path`.

Key: `(inspection_id, valid_from)`.

Example, IN-2041187:

valid_from	valid_to	record_status	classification	batch
2026-10-05	2026-10-12	ACTIVE	VAI	snapshot 10-05
2026-10-12	9999-12-31	ACTIVE	OAI	snapshot 10-12

Example, IN-2040551 (vanished from the 10-12 file):

valid_from	valid_to	record_status	classification
2026-10-05	2026-10-12	ACTIVE	OAI
2026-10-12	9999-12-31	RETRACTED	OAI (kept)

fact_gmp_document — S2, EU GMP certificates and non-compliance (SCD2)

Column	Description
doc_ref	EU's ID, e.g. EUGMP–NCR–2026–0412. Natural key.
valid_from, valid_to, is_current, record_status, batch_id	As in fact_inspection.
doc_type	GMP_CERTIFICATE or NON_COMPLIANCE.
site_ref	EU's site identifier → dim_site_identifier.
site_name_as_published, address_as_published	Verbatim, UTF-8 (non-ASCII preserved).
country_code	ISO-2.
issuing_authority	e.g. ANSM (FR).
inspection_end_date, issued_date	Parsed from DD/MM/YYYY.
status	ACTIVE or WITHDRAWN (agency's own status, distinct from our record_status).

Note: duplicates inside one export are removed before loading (same doc_ref, same content).

fact_warning_letter — S3 (SCD2)

Column	Description
letter_id	e.g. WL–320–26–118. Natural key.
valid_from, valid_to, is_current, record_status, batch_id	As above.
fei_number	Usually NULL. When NULL the letter is matched by name/address and gets its own LETTER_ID row in the bridge.
recipient_firm_as_published, recipient_address_as_published	Verbatim.
country_code	ISO-2 (derived from the address).
issue_date	Parsed.
subject	As published.
body_s3_path	Letter text stays in S3 (5–60 KB each); only the pointer is here.
source_url	Agency URL.
nlp_status	PENDING / DONE / FAILED — progress of the external tagging call.

fact_letter_tag — NLP output

Column	Description
letter_id	→ fact_warning_letter.
tag	e.g. DATA_INTEGRITY.
confidence	From the NLP service.
nlp_model_version	Kept so a re-tag adds rows instead of replacing them.
tagged_at	When.

Key: `(letter_id, nlp_model_version, tag)`.

fact_enforcement_action — S4 (versioned by the agency)

The agency sends a `version` with every event, so we do not need `valid_to`: **current state = highest version**. If the highest version is a `RETRACT`, the action is retracted.

Column	Description
`action_id`	e.g. `IA-66-40-0098`.
`action_version`	From the event. Key with `action_id`.
`event_id`	The stream event that produced this row, for traceability to the S3 event archive.
`operation`	`UPSERT` or `RETRACT`.
`action_type`	`IMPORT_ALERT` / `RECALL` / `DEBARMENT`.
`agency`	e.g. `FDA`.
`fei_number`	→ `dim_site_identifier`. NULL on a bare RETRACT.
`firm_name`	Schema v1 field `firm` and v2 field `firm_name` both land here.
`country_code`	ISO-2.
`product_codes`	JSON list; v2 only.
`effective_date`	Agency's date.
`event_time`	Agency's timestamp on the event.
`loaded_at`	When we applied it. Used for "what did we show on date X".

Key: `(action_id, action_version)`.

Applying the four sample events in arrival order:

event	result
e-88f1, v1 UPSERT	insert (IA-66-40-0098, 1). Current = v1. Alert.
e-91c3, v2 UPSERT	insert (IA-66-40-0098, 2). Current = v2. Alert (changed action).
e-88f1 again	rejected: `(IA-66-40-0098, 1)` already exists (primary key). Nothing happens.
e-93aa, v3 RETRACT for RC-2026-1187	insert (RC-2026-1187, 3, RETRACT). We've never seen v1/v2 of it; the row acts as a tombstone so a late v1 cannot become current. No FEI, so no alert; flag for review.

dim_customer

Column	Description
`customer_id`	e.g. `cust_042`.
`customer_name`	Display name.

dim_watchlist_entry — S5 (SCD2 from CDC)

Column	Description
watch_id	From the customer app, e.g. 9912. Natural key.
customer_id	→ dim_customer. **Row-level security filters on this.**
customer_supplier_ref	Customer's own supplier code, e.g. SUP-0071.
supplier_name_as_typed, country_as_typed, fei_number_as_typed	Free text exactly as the customer entered it.
is_active	false after a CDC delete. A delete must stop alerts immediately, so this flag is also pushed to the alert lookup at once.
valid_from, valid_to, is_current	SCD2 (customers edit supplier names).
cdc_timestamp	ts_ms from the CDC event; older events are ignored (out-of-order guard).

Reaches its site through dim_site_identifier where identifier_type = 'WATCH_ID'.

fact_alert

Column	Description
alert_id	Generated.
customer_id, watch_id	Who is being alerted and why. RLS applies.
site_key	The site at the time of the alert.
action_id, action_version	Which enforcement action version triggered it.
created_at, delivered_at	Timeline.

Unique: (watch_id, action_id, action_version) so the same event can never alert twice.

match_review_queue — U6 work queue

Column	Description
identifier_type, identifier_value	The thing we couldn't confidently place.
candidate_sites	JSON list of {site_key, score, evidence}.
status	OPEN / LINKED / NEW_SITE / REJECTED.
opened_at, reviewed_by, reviewed_at	Timeline.

ingest_batch — lineage

Column	Description
`batch_id`	Generated. Every S1/S2/S3 fact version points at one of these.
`source_code`	`S1`, `S2` or `S3`.
`received_at`	When the file arrived. Becomes `valid_from` on the fact versions it introduces.
`s3_path`	The immutable raw file. This is the "traceable to its source record" link for U1 and the replay input.
`row_count`, `status`	`RECEIVED` / `PARSED` / `LOADED` / `FAILED` / `QUARANTINED`.
`parser_version`	Which parser produced the load, so a reprocess is distinguishable from the original.
`dq_checks`	JSON of the data-quality results for this file (row count vs last week, null rates, date parse failures).

Stream events (S4, S5) are not tracked here: duplicates are rejected by the fact table's primary key, and replay reads the raw event archive in S3.

Appendix B — Full DDL

```
-- Part A.2 — Schema definitions (PostgreSQL 15+)
-- System of record for the Supplier Watch model. Column meanings: see 01_entity_model.md.
--
-- Conventions
--   * fact_ tables for S1/S2/S3 are SCD Type 2: (natural_key, valid_from) is the key,
--     valid_to = 'infinity' while current, is_current mirrors that for easy filtering.
--     Rows are never UPDATEd except to close valid_to / flip is_current.
--   * valid_from / valid_to mean "when OUR PLATFORM showed this version", not agency dates.
--   * *_as_published columns keep the agency's text verbatim; normalised columns may be
--     recomputed on reprocessing without touching raw files.

CREATE EXTENSION IF NOT EXISTS btree_gist;   -- non-overlap constraints on SCD2 ranges
CREATE EXTENSION IF NOT EXISTS pg_trgm;      -- name/address similarity for matching candidates

-- ============================================================================
-- LINEAGE
-- ============================================================================
-- One row per batch file received (S1/S2/S3). Stream events are not tracked here: duplicates are
-- rejected by the fact table's primary key and replay reads the raw event archive in S3.
CREATE TABLE ingest_batch (
    batch_id        uuid PRIMARY KEY,
    source_code     text NOT NULL CHECK (source_code IN ('S1','S2','S3')),
    received_at     timestamptz NOT NULL,                  -- becomes valid_from on the fact
versions it introduces
    s3_path         text NOT NULL,                         -- immutable raw file
    row_count       int,
    status          text NOT NULL CHECK (status IN
('RECEIVED','PARSED','LOADED','FAILED','QUARANTINED')),
    parser_version  text NOT NULL,
    dq_checks       jsonb
);

-- ============================================================================
-- DIMENSIONS
-- ============================================================================
CREATE TABLE dim_site (
    site_key      text PRIMARY KEY,                        -- 'S-000001', generated by us
    site_name     text NOT NULL,
    country_code  char(2),
    city          text,
    address       text,
    is_active     boolean NOT NULL DEFAULT true,           -- false once merged into another site
    created_at    timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX dim_site_name_trgm    ON dim_site USING gin (site_name gin_trgm_ops);
CREATE INDEX dim_site_address_trgm ON dim_site USING gin (address gin_trgm_ops);
CREATE INDEX dim_site_block        ON dim_site (country_code, city);     -- blocking key for
matching

-- The bridge: agency identifier -> our site, as SCD2.
-- Steward actions (merge, split, link, unlink) are recorded on the rows they open/close:
-- every row touched by one action shares a decision_group_id, so a split can find and reverse a
merge.
CREATE TABLE dim_site_identifier (
    identifier_type         text NOT NULL CHECK (identifier_type IN
('FEI','EU_SITE_REF','LETTER_ID','WATCH_ID')),
    identifier_value        text NOT NULL,
    valid_from              timestamptz NOT NULL DEFAULT now(),
    valid_to                timestamptz NOT NULL DEFAULT 'infinity',
```

```
    is_current              boolean NOT NULL DEFAULT true,
    site_key                text NOT NULL REFERENCES dim_site,
    match_method            text NOT NULL CHECK (match_method IN
('EXACT_ID','RULE','FUZZY','STEWARD')),
    match_score             numeric(4,3),
    match_evidence          jsonb,
    matching_rules_version  text NOT NULL,
    decided_by              text NOT NULL,                -- steward user id, or 'system:er-v3'
    decision_reason         text,                         -- free text; NULL for automatic
matches
    decision_group_id       uuid,                         -- shared by all rows one steward
action touched
    PRIMARY KEY (identifier_type, identifier_value, valid_from),
    -- one identifier belongs to exactly one site at any instant
    EXCLUDE USING gist (identifier_type WITH =, identifier_value WITH =, tstzrange(valid_from,
valid_to) WITH &&)
);
CREATE INDEX dim_site_identifier_current_by_id   ON dim_site_identifier (identifier_type,
identifier_value) WHERE is_current;
CREATE INDEX dim_site_identifier_current_by_site ON dim_site_identifier (site_key) WHERE
is_current;

CREATE TABLE match_review_queue (
    review_id         uuid PRIMARY KEY,
    identifier_type   text NOT NULL,
    identifier_value  text NOT NULL,
    candidate_sites   jsonb NOT NULL,                     -- [{site_key, score, evidence}, ...]
    status            text NOT NULL DEFAULT 'OPEN' CHECK (status IN
('OPEN','LINKED','NEW_SITE','REJECTED')),
    opened_at         timestamptz NOT NULL DEFAULT now(),
    reviewed_by       text,
    reviewed_at       timestamptz,
    decision_group_id uuid                                -- the bridge rows the review produced
);
CREATE UNIQUE INDEX match_review_queue_open ON match_review_queue (identifier_type,
identifier_value) WHERE status = 'OPEN';

CREATE TABLE dim_customer (
    customer_id    text PRIMARY KEY,
    customer_name  text NOT NULL
);

CREATE TABLE dim_watchlist_entry (
    watch_id               bigint NOT NULL,               -- from the customer app
    valid_from             timestamptz NOT NULL,
    valid_to               timestamptz NOT NULL DEFAULT 'infinity',
    is_current             boolean NOT NULL DEFAULT true,
    customer_id            text NOT NULL REFERENCES dim_customer,
    customer_supplier_ref  text,
    supplier_name_as_typed text NOT NULL,
    country_as_typed       text,
    fei_number_as_typed    text,
    is_active              boolean NOT NULL,              -- false after a CDC delete
    cdc_timestamp          bigint NOT NULL,               -- ts_ms; older events are ignored
    PRIMARY KEY (watch_id, valid_from),
    EXCLUDE USING gist (watch_id WITH =, tstzrange(valid_from, valid_to) WITH &&)
);
CREATE INDEX dim_watchlist_entry_by_customer ON dim_watchlist_entry (customer_id) WHERE
is_current AND is_active;

-- ============================================================================
-- FACTS
-- ============================================================================

-- S1: FDA inspection classifications. Weekly full snapshot; versions come from diffing
snapshots.
```

```
CREATE TABLE fact_inspection (
    inspection_id                    text NOT NULL,
    valid_from                       timestamptz NOT NULL,
    valid_to                         timestamptz NOT NULL DEFAULT 'infinity',
    is_current                       boolean NOT NULL DEFAULT true,
    record_status                    text NOT NULL CHECK (record_status IN
('ACTIVE','RETRACTED')),
    batch_id                         uuid NOT NULL REFERENCES ingest_batch,
    fei_number                       text NOT NULL,
    firm_name_as_published           text,
    city                             text,
    country_code                     char(2),
    inspection_end_date              date,
    inspection_end_date_as_published  text,
    classification                   text CHECK (classification IN ('NAI','VAI','OAI')),
    product_type                     text,
    published_date                   date,
    PRIMARY KEY (inspection_id, valid_from),
    EXCLUDE USING gist (inspection_id WITH =, tstzrange(valid_from, valid_to) WITH &&)
);
CREATE INDEX fact_inspection_current_by_fei ON fact_inspection (fei_number) WHERE is_current;
CREATE INDEX fact_inspection_by_end_date    ON fact_inspection (inspection_end_date);

-- S2: EU GMP certificates and non-compliance reports. Monthly export; same diff-based SCD2.
CREATE TABLE fact_gmp_document (
    doc_ref                  text NOT NULL,
    valid_from               timestamptz NOT NULL,
    valid_to                 timestamptz NOT NULL DEFAULT 'infinity',
    is_current               boolean NOT NULL DEFAULT true,
    record_status            text NOT NULL CHECK (record_status IN ('ACTIVE','RETRACTED')),
    batch_id                 uuid NOT NULL REFERENCES ingest_batch,
    doc_type                 text CHECK (doc_type IN ('GMP_CERTIFICATE','NON_COMPLIANCE')),
    site_ref                 text NOT NULL,
    site_name_as_published   text,
    address_as_published     text,
    country_code             char(2),
    issuing_authority        text,
    inspection_end_date      date,
    issued_date              date,
    status                   text CHECK (status IN ('ACTIVE','WITHDRAWN')),   -- the agency's
status
    PRIMARY KEY (doc_ref, valid_from),
    EXCLUDE USING gist (doc_ref WITH =, tstzrange(valid_from, valid_to) WITH &&)
);
CREATE INDEX fact_gmp_document_current_by_site_ref ON fact_gmp_document (site_ref) WHERE
is_current;

-- S3: Warning letters. Body text stays in S3.
CREATE TABLE fact_warning_letter (
    letter_id                       text NOT NULL,
    valid_from                      timestamptz NOT NULL,
    valid_to                        timestamptz NOT NULL DEFAULT 'infinity',
    is_current                      boolean NOT NULL DEFAULT true,
    record_status                   text NOT NULL CHECK (record_status IN
('ACTIVE','RETRACTED')),
    batch_id                        uuid NOT NULL REFERENCES ingest_batch,
    fei_number                      text,                -- usually NULL
    recipient_firm_as_published     text,
    recipient_address_as_published  text,
    country_code                    char(2),
    issue_date                      date,
    subject                         text,
    body_s3_path                    text NOT NULL,
    source_url                      text,
    nlp_status                      text NOT NULL DEFAULT 'PENDING' CHECK (nlp_status IN
('PENDING','DONE','FAILED')),
```

```
    PRIMARY KEY (letter_id, valid_from),
    EXCLUDE USING gist (letter_id WITH =, tstzrange(valid_from, valid_to) WITH &&)
);

CREATE TABLE fact_letter_tag (
    letter_id          text NOT NULL,
    nlp_model_version  text NOT NULL,
    tag                text NOT NULL,
    confidence         numeric(4,3),
    tagged_at          timestamptz NOT NULL DEFAULT now(),
    PRIMARY KEY (letter_id, nlp_model_version, tag)
);

-- S4: Enforcement actions. Versioned by the agency; current state = highest action_version.
CREATE TABLE fact_enforcement_action (
    action_id       text NOT NULL,
    action_version  int  NOT NULL,
    event_id        text NOT NULL,                        -- traceability to the S3 event
archive
    operation       text NOT NULL CHECK (operation IN ('UPSERT','RETRACT')),
    action_type     text,
    agency          text,
    fei_number      text,                                 -- NULL on a bare RETRACT
    firm_name       text,                                 -- v1 'firm' and v2 'firm_name' both
land here
    country_code    char(2),
    product_codes   jsonb,                                -- v2 only
    effective_date  date,
    event_time      timestamptz NOT NULL,
    loaded_at       timestamptz NOT NULL DEFAULT now(),
    PRIMARY KEY (action_id, action_version)
);
CREATE INDEX fact_enforcement_action_by_fei ON fact_enforcement_action (fei_number);

CREATE TABLE fact_alert (
    alert_id        uuid PRIMARY KEY,
    customer_id     text   NOT NULL REFERENCES dim_customer,
    watch_id        bigint NOT NULL,
    site_key        text   NOT NULL REFERENCES dim_site,
    action_id       text   NOT NULL,
    action_version  int    NOT NULL,
    created_at      timestamptz NOT NULL DEFAULT now(),
    delivered_at    timestamptz,
    UNIQUE (watch_id, action_id, action_version)          -- idempotent fan-out
);

-- ============================================================================
-- TENANT ISOLATION (enforced in the database, not only in app code)
-- ============================================================================
ALTER TABLE dim_watchlist_entry ENABLE ROW LEVEL SECURITY;
ALTER TABLE fact_alert          ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_watchlist ON dim_watchlist_entry USING (customer_id =
current_setting('app.customer_id', true));
CREATE POLICY tenant_alert     ON fact_alert          USING (customer_id =
current_setting('app.customer_id', true));
-- pipeline roles are BYPASSRLS; the app role sets app.customer_id per connection.

-- ============================================================================
-- CURRENT-STATE VIEWS
-- ============================================================================
CREATE VIEW v_inspection_current AS
    SELECT * FROM fact_inspection WHERE is_current AND record_status = 'ACTIVE';

CREATE VIEW v_gmp_document_current AS
    SELECT * FROM fact_gmp_document WHERE is_current AND record_status = 'ACTIVE';
```

```
CREATE VIEW v_warning_letter_current AS
    SELECT * FROM fact_warning_letter WHERE is_current AND record_status = 'ACTIVE';

CREATE VIEW v_enforcement_action_current AS
    SELECT DISTINCT ON (action_id) *
    FROM fact_enforcement_action
    ORDER BY action_id, action_version DESC;               -- retracted actions have operation =
'RETRACT'

CREATE VIEW v_site_identifier_current AS
    SELECT * FROM dim_site_identifier WHERE is_current;

-- U1: every current fact for a site, one row each, with its source id and batch (traceability).
CREATE VIEW v_site_facts AS
    SELECT b.site_key, 'INSPECTION' AS fact_type, i.inspection_id AS fact_id,
i.inspection_end_date AS fact_date,
           i.classification AS outcome, 'FDA' AS agency, i.batch_id
      FROM v_site_identifier_current b
      JOIN v_inspection_current i ON b.identifier_type = 'FEI' AND b.identifier_value =
i.fei_number
    UNION ALL
    SELECT b.site_key, 'GMP_DOCUMENT', g.doc_ref, g.issued_date, g.status, g.issuing_authority,
g.batch_id
      FROM v_site_identifier_current b
      JOIN v_gmp_document_current g ON b.identifier_type = 'EU_SITE_REF' AND b.identifier_value =
g.site_ref
    UNION ALL
    SELECT b.site_key, 'WARNING_LETTER', w.letter_id, w.issue_date, w.subject, 'FDA', w.batch_id
      FROM v_site_identifier_current b
      JOIN v_warning_letter_current w ON b.identifier_type = 'LETTER_ID' AND b.identifier_value =
w.letter_id
    UNION ALL
    SELECT b.site_key, 'ENFORCEMENT_ACTION', e.action_id, e.effective_date, e.action_type,
e.agency, NULL
      FROM v_site_identifier_current b
      JOIN v_enforcement_action_current e ON b.identifier_type = 'FEI' AND b.identifier_value =
e.fei_number
     WHERE e.operation <> 'RETRACT';

-- ============================================================================
-- Scale notes
-- ============================================================================
-- Today: ~250k inspections + 60k EU docs + 15y history = low millions of SCD2 rows. No
partitioning;
-- the partial indexes on is_current keep "current" reads small.
-- At 10x (25 agencies, 1M sites): LIST-partition fact tables by agency so a re-parse of one
agency
-- rewrites one partition; move U4 joins to the Snowflake star.
--
-- If Snowflake were the system of record: same tables, but PK/FK/EXCLUDE are not enforced,
-- partial indexes become CLUSTER BY, RLS becomes row access policies, and merge/split loses
-- single-transaction atomicity.
```

Appendix C — Sample data as loaded

```
-- Sample data from the brief, loaded as the pipeline would leave it after the 2026-10-12 S1
snapshot.
-- Used to prove the U2/U3/U4 queries. All data synthetic (from the brief).

-- ------------------------------------------------------------------ lineage
INSERT INTO ingest_batch (batch_id, source_code, received_at, s3_path, row_count, status,
parser_version) VALUES
 ('00000000-0000-0000-0000-000000000105','S1','2026-10-05
06:00+00','s3://raw/s1/received_date=2026-10-05/fda_inspections.csv',3,'LOADED','s1-parser-1.0'),
 ('00000000-0000-0000-0000-000000000112','S1','2026-10-12
06:00+00','s3://raw/s1/received_date=2026-10-12/fda_inspections.csv',2,'LOADED','s1-parser-1.0'),
 ('00000000-0000-0000-0000-000000000201','S2','2026-10-01
03:00+00','s3://raw/s2/received_date=2026-10-01/eu_gmp.json',1,'LOADED','s2-parser-1.0'),
 ('00000000-0000-0000-0000-000000000302','S3','2026-10-02
09:00+00','s3://raw/s3/received_date=2026-10-02/WL-320-26-118.json',1,'LOADED','s3-parser-1.0');

-- ------------------------------------------------------------------ our factories
INSERT INTO dim_site (site_key, site_name, country_code, city, address) VALUES
 ('S-000001','Acme Pharma Unit II, Sangareddy','IN','hyderabad','plot 14 ida pashamylaram
sangareddy telangana 502307'),
 ('S-000002','Acme Pharma Unit I, Visakhapatnam','IN','visakhapatnam',NULL);

-- ------------------------------------------------------------------ bridge
INSERT INTO dim_site_identifier
 (identifier_type, identifier_value, valid_from, site_key, match_method, match_score,
match_evidence, matching_rules_version, decided_by) VALUES
 ('FEI','3004521877','2026-10-05 06:05+00','S-000001','EXACT_ID',1.0,'{"rule":"same FEI"}','er-
1.0','system:er-1.0'),
 ('FEI','3009876543','2026-10-05 06:05+00','S-000002','EXACT_ID',1.0,'{"rule":"same FEI"}','er-
1.0','system:er-1.0'),
 ('EU_SITE_REF','LOC-118842','2026-10-01 03:05+00','S-000001','FUZZY',0.91,
    '{"name_sim":0.82,"address_sim":0.95,"same_country":true,"same_city_block":true}','er-
1.0','system:er-1.0'),
 ('LETTER_ID','WL-320-26-118','2026-10-03 10:00+00','S-000001','STEWARD',1.0,
    '{"address_sim":0.93,"note":"Plot 14 IDA Pashamylaram matches EU record"}','er-
1.0','steward:priya'),
 ('WATCH_ID','9912','2026-10-14 10:00+00','S-000001','FUZZY',0.87,
    '{"name_sim":0.84,"same_country":true,"city_in_name":"hyderabad"}','er-1.0','system:er-1.0');

-- ------------------------------------------------------------------ S1 inspections (two snapshots
diffed)
-- Snapshot 10-05 introduced three rows. Snapshot 10-12: IN-2041187 VAI→OAI (CORRECT), IN-2040551
gone (RETRACT).
INSERT INTO fact_inspection
 (inspection_id, valid_from, valid_to, is_current, record_status, batch_id, fei_number,
firm_name_as_published, city, country_code,
  inspection_end_date, inspection_end_date_as_published, classification, product_type,
published_date) VALUES
 ('IN-2041187','2026-10-05 06:00+00','2026-10-12 06:00+00',false,'ACTIVE','00000000-0000-0000-
0000-000000000105',
   '3004521877','Acme Pharma Pvt. Ltd.','Hyderabad','IN','2026-08-14','2026-08-
14','VAI','Drugs','2026-09-30'),
 ('IN-2041187','2026-10-12 06:00+00','infinity',true,'ACTIVE','00000000-0000-0000-0000-
000000000112',
   '3004521877','Acme Pharma Pvt. Ltd.','Hyderabad','IN','2026-08-14','2026-08-
14','OAI','Drugs','2026-09-30'),
 ('IN-2039902','2026-10-05 06:00+00','infinity',true,'ACTIVE','00000000-0000-0000-0000-
000000000105',
   '3004521877','ACME PHARMACEUTICALS PRIVATE LIMITED - UNIT II','Hyderabad','IN','2026-08-
```

```
02','08/02/2026','NAI','Drugs','2026-08-28'),
 ('IN-2040551','2026-10-05 06:00+00','2026-10-12 06:00+00',false,'ACTIVE','00000000-0000-0000-
0000-000000000105',
   '3009876543','Acme Pharma Pvt Ltd - Unit I','Visakhapatnam','IN','2026-07-30','2026-07-
30','OAI','Drugs','2026-09-12'),
 ('IN-2040551','2026-10-12 06:00+00','infinity',true,'RETRACTED','00000000-0000-0000-0000-
000000000112',
   '3009876543','Acme Pharma Pvt Ltd - Unit I','Visakhapatnam','IN','2026-07-30','2026-07-
30','OAI','Drugs','2026-09-12');

-- ---------------------------------------------------------------- S2
INSERT INTO fact_gmp_document
 (doc_ref, valid_from, record_status, batch_id, doc_type, site_ref, site_name_as_published,
address_as_published, country_code,
  issuing_authority, inspection_end_date, issued_date, status) VALUES
 ('EUGMP-NCR-2026-0412','2026-10-01 03:00+00','ACTIVE','00000000-0000-0000-0000-
000000000201','NON_COMPLIANCE','LOC-118842',
  'Acme Pharma Private Ltd, Unit 2','Plot 14, IDA Pashamylaram, Sangareddy, Telangana
502307','IN','ANSM (FR)','2026-07-21','2026-09-03','ACTIVE');

-- ---------------------------------------------------------------- S3
INSERT INTO fact_warning_letter
 (letter_id, valid_from, record_status, batch_id, fei_number, recipient_firm_as_published,
recipient_address_as_published, country_code,
  issue_date, subject, body_s3_path, source_url, nlp_status) VALUES
 ('WL-320-26-118','2026-10-02 09:00+00','ACTIVE','00000000-0000-0000-0000-
000000000302',NULL,'Acme Pharma Pvt Ltd',
  'Plot No. 14, IDA Pashamylaram, Sangareddy Dist., Telangana, India','IN','2026-10-
02','CGMP/Finished Pharmaceuticals/Adulterated',
  's3://raw/s3/received_date=2026-10-02/WL-320-26-118.txt','https://example.gov/wl/320-26-
118','DONE');

INSERT INTO fact_letter_tag (letter_id, nlp_model_version, tag, confidence) VALUES
 ('WL-320-26-118','nlp-2.1','DATA_INTEGRITY',0.94),
 ('WL-320-26-118','nlp-2.1','CGMP_DEVIATION',0.88),
 ('WL-320-26-118','nlp-2.1','ADULTERATION',0.71);

-- ---------------------------------------------------------------- S4: the four events applied
in arrival order
-- e-88f1 v1, e-91c3 v2, e-88f1 again (rejected by PK), e-93aa v3 RETRACT of an unseen action.
INSERT INTO fact_enforcement_action
 (action_id, action_version, event_id, operation, action_type, agency, fei_number, firm_name,
country_code, product_codes, effective_date, event_time, loaded_at) VALUES
 ('IA-66-40-0098',1,'e-88f1','UPSERT','IMPORT_ALERT','FDA','3004521877','Acme Pharma Pvt
Ltd','IN',NULL,'2026-10-11','2026-10-11 14:02:10+00','2026-10-11 14:02:30+00'),
 ('IA-66-40-0098',2,'e-91c3','UPSERT','IMPORT_ALERT','FDA','3004521877','Acme Pharma Pvt
Ltd','IN','["62GDY99"]','2026-10-11','2026-10-12 09:30:00+00','2026-10-12 09:30:20+00'),
 ('RC-2026-1187',3,'e-93aa','RETRACT','RECALL','FDA',NULL,NULL,NULL,NULL,NULL,'2026-10-13
08:00:00+00','2026-10-13 08:00:15+00');
-- the duplicate e-88f1 must be a no-op:
INSERT INTO fact_enforcement_action
 (action_id, action_version, event_id, operation, action_type, agency, fei_number, firm_name,
country_code, effective_date, event_time) VALUES
 ('IA-66-40-0098',1,'e-88f1','UPSERT','IMPORT_ALERT','FDA','3004521877','Acme Pharma Pvt
Ltd','IN','2026-10-11','2026-10-11 14:02:10+00')
ON CONFLICT (action_id, action_version) DO NOTHING;

-- ---------------------------------------------------------------- S5
INSERT INTO dim_customer VALUES ('cust_042','Sample Drug Maker GmbH');

INSERT INTO dim_watchlist_entry
 (watch_id, valid_from, customer_id, customer_supplier_ref, supplier_name_as_typed,
country_as_typed, fei_number_as_typed, is_active, cdc_timestamp) VALUES
 (9912,'2026-10-14 09:20+00','cust_042','SUP-0071','Acme Pharma Unit-II
Hyderabad','India',NULL,true,1760436000000);
```

Appendix D — Query results (PostgreSQL 17)

```
 customer_id
 cust_042
(1 row)
 inspection_id | classification | record_status |        valid_from         |         valid_to
|               batch_id
 IN-2041187    | VAI            | ACTIVE        | 2026-10-05 11:30:00+05:30 | 2026-10-12
11:30:00+05:30 | 00000000-0000-0000-0000-000000000105
(1 row)
 site_key |            site_name            | match_method
 S-000001 | Acme Pharma Unit II, Sangareddy | EXACT_ID
(1 row)
 country_code |  quarter   | inspections_all | oai_rate_all_pct | inspections_watchlist |
oai_rate_watchlist_pct
 IN           | 2026-07-01 |               2 |             50.0 |                     2 |
50.0
(1 row)

--- U3 again for 2026-10-13 (after the correction) ---
 classification | record_status
----------------+---------------
 OAI            | ACTIVE
(1 row)

--- U3 for the retracted one, as of 10-08 vs today ---
   as_of    | classification | record_status
------------+----------------+---------------
 2026-10-08 | OAI            | ACTIVE
 today      | OAI            | RETRACTED
(2 rows)

--- S4: current state per action (duplicate e-88f1 was a no-op: 3 rows total) ---
   action_id   | action_version | operation | event_id | fei_number
---------------+----------------+-----------+----------+------------
 IA-66-40-0098 |              2 | UPSERT    | e-91c3   | 3004521877
 RC-2026-1187  |              3 | RETRACT   | e-93aa   |
(2 rows)

--- U1: everything we know about S-000001, each row traceable to its source ---
     fact_type      |       fact_id       | fact_date  |                  outcome
|  agency   |               batch_id
--------------------+---------------------+------------+-----------------------------------------
--+-----------+--------------------------------------
 INSPECTION         | IN-2039902          | 2026-08-02 | NAI
| FDA       | 00000000-0000-0000-0000-000000000105
 INSPECTION         | IN-2041187          | 2026-08-14 | OAI
| FDA       | 00000000-0000-0000-0000-000000000112
 GMP_DOCUMENT       | EUGMP-NCR-2026-0412 | 2026-09-03 | ACTIVE
| ANSM (FR) | 00000000-0000-0000-0000-000000000201
 WARNING_LETTER     | WL-320-26-118       | 2026-10-02 | CGMP/Finished
Pharmaceuticals/Adulterated | FDA       | 00000000-0000-0000-0000-000000000302
 ENFORCEMENT_ACTION | IA-66-40-0098       | 2026-10-11 | IMPORT_ALERT
| FDA       |
(5 rows)
```